

HANZLA NAWAZ

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Summary

Machine Learning Engineer with expertise in developing LLM-powered applications for automating workflows and generating customer insights. Achieved significant advancements by refining open-source LLMs for healthcare NLP and optimizing data preprocessing to enhance public health forecasting. Known for architecting robust strategies in AI application development and optimizing machine learning solutions to increase operational efficiency. Eager to apply experience to create cutting-edge machine learning systems that deliver transformative data insights.

EDUCATION

SUPERIOR UNIVERSITY

Bachelor of Science, Artificial Intelligence

Jul 2024

- **GPA:** 3.30/4.0

TECHNICAL SKILLS

- **Programming & Frameworks:** Python, JavaScript, SQL, NoSQL, HTML/CSS, FastAPI, Flask, Beautiful Soup, Gradio, REST APIs
- **Machine Learning & AI:** TensorFlow, Keras, Scikit learn, NumPy, Pandas, Matplotlib, Seaborn, Plotly, SciPy, CNN, RNN, LSTM, GPT, BERT, Transfer Learning, Large Language Models, Fine Tuning, Hugging Face Transformers, OpenAI GPT, LangChain, RAG, LangGraph, CrewAI
- **Data & Visualization Tools:** Jupyter Notebooks, Power BI, Tableau, Excel, PowerPoint
- **DevOps, MLOps & Cloud:** Docker, MLflow, CI/CD (GitHub Actions, DVC), Git, AWS, GCP, PostgreSQL, MongoDB, MS SQL Server, Pinecone, Qdrant
- **Soft Skills:** Problem Solving, Communication, Leadership, Collaboration, Project Management

WORK EXPERIENCE

XEVEN SOLUTIONS (Pvt) Ltd.

Machine Learning Engineer

Jul 2024 - Present

Johar Town, Lahore

- Created LLM-powered applications using OpenAI APIs to automate workflows and speed up customer insights creation, integrating A2A principles to enhance data interaction while maintaining ethical design standards, resulting in improved operational efficiency
- Fine-tuned open-source LLMs (using QLoRA, LoRA, and Unsloth) for healthcare-specific NLP tasks, optimizing inference speed and domain performance for production-ready pipelines through Agile methodologies and full software lifecycle management.
- Developed API integration strategies for AI-powered applications across education, recruitment, and marketing sectors utilizing LangChain, LangGraph, OpenAI APIs, and open-source models; implemented RAG, prompt engineering, FastAPI deployment, and event driven workflows involving async processes.
- Architected PostgreSQL and MS SQL Server schemas for application logging and bot history management, focusing on architecting reliable data systems while emphasizing interoperability of data sources.
- Collaborated closely with data scientists, software engineers, and stakeholders to align ML solutions with business objectives, ensuring ethical practices in delivering solutions, which led to more aligned and ethical ML implementations

OMDENA

Machine Learning Engineer

Nov 2023 - May 2024

Remote

- Automated preprocessing of tuberculosis case data in Kaduna, Nigeria, reducing manual data-cleaning effort by 50% through effective data processing and ETL pipeline design, focusing on data decomposition for better clarity.
- Developed predictive regression models for TB case outcomes, improving forecast accuracy by 15% to support public health interventions by utilizing best practices in data science, leveraging concepts from data engineering principles.
- Collaborated with global teams to counter misinformation in Nepal, leveraging AI tools (CrewAI, Groq API) to identify and flag false content, while maintaining ethical standards in data usage and adhering to ethics guidelines.
- Implemented data validation and quality assurance protocols for healthcare datasets, contributing to continuous improvement in debugging and data integrity while utilizing a loop process for ongoing evaluations.

PROJECTS

AI-Powered HR Recruitment & Candidate Screening Platform

- Developed a comprehensive AI-driven recruitment system that automates candidate ingestion, semantic matching, CV analysis, and multi-source data integration using vector embeddings, machine learning, and scalable cloud microservices to streamline HR workflows and boost hiring efficiency.

MediStreamline AI – Intelligent Medical Consultation Platform

- Developed and deployed a comprehensive, AI-driven medical consultation system using FastAPI, OpenAI GPT4/Mistral AI, and Google Cloud services.
- The platform delivers multilingual diagnosis, realtime speeche totext/TexttoSpeech, medical image analysis, automated report generation, and robust cloud/DevOps infrastructure to support healthcare professionals with sub-200 ms API response times and 99.9% uptime; utilized containerization, such as Docker and Kubernetes, to enhance application scalability and resilience.